FinFET & FD-SOI Technologies

[116] Design of VLSI Systems

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FinFET

- **Technology overview**
- "Comprehensive Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications", K. Karimi et al. [1]
- "FinFET to GAA MBCFET: A Review and Insights", R. R. Das et al. [2]

Introduction to FinFET Technology

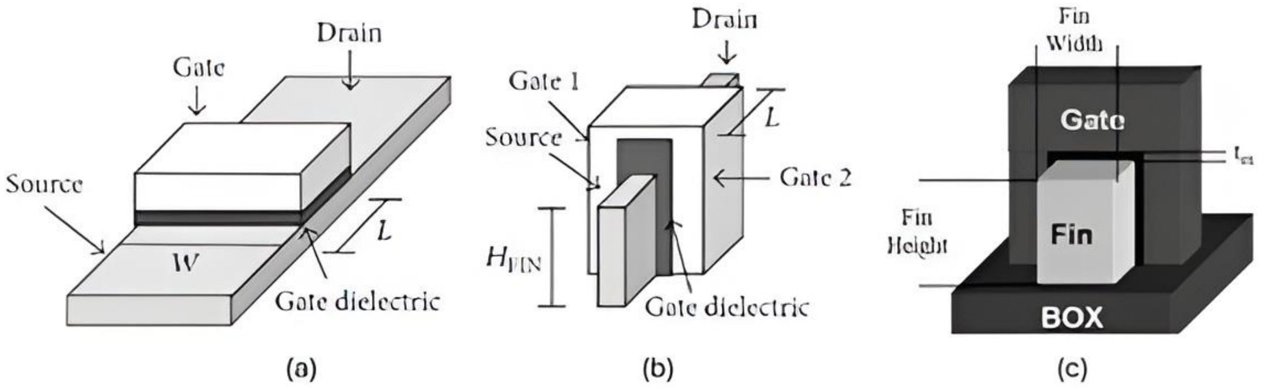

Figure FinFET.1: Structure Schematic of (a) planar MOSFET and (b) FinFET. (c) Demonstrating H_{Fin} as fin height and W_{Fin} as fin width. Figure taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

- FinFET = Three-dimensional multi-gate MOSFET
- Developed to overcome short-channel effects in planar CMOS
- Introduced ~2000 by C. Hu (Berkeley)
- Enables continued CMOS scaling below 32 nm

Motivation: Limitations of Planar MOSFETs

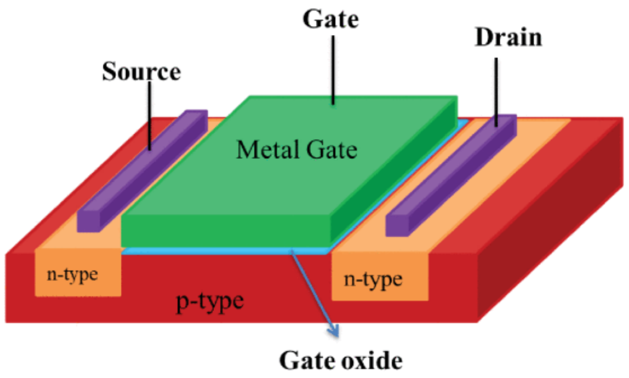

Figure FinFET.2: The planar MOSFET semiconductor device.

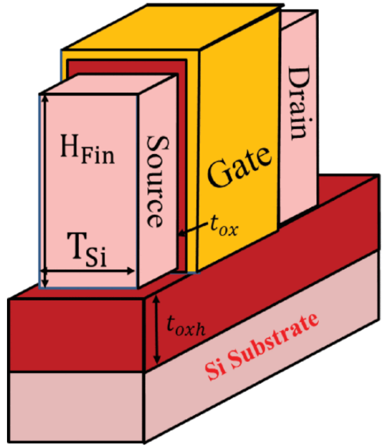

Figure FinFET.3: Schematic diagram of conventional FinFET.

Figures taken from R. R. Das, T. R. Rajalekshmi and A. James, "FinFET to GAA MBCFET: A Review and Insights," in *IEEE Access*, vol. 12, pp. 50556-50577, 2024, doi: 10.1109/ACCESS.2024.3384428.

- At < 32nm: strong SCE, leakage currents ↑, threshold voltage variation ↑
- Gate control weak → poor electrostatics and power inefficiency
- Need for multi-gate structures (MuGFETs)

Historical Development

- **1904** – Vaccum Tube
- **1947** – First Point Contact Transistor
- **1948** – Invention of Bipolar Junction Transistor
- 1956 – Design of Insulated-Gate Field Effect Transistor (aka MOSFET)
- **1958** – **Development of First Integrated Circuit**
- **1987** – **First Multi-Gate Transistor**
- **1989** – **First Double-Gate Transistor named DELTA**

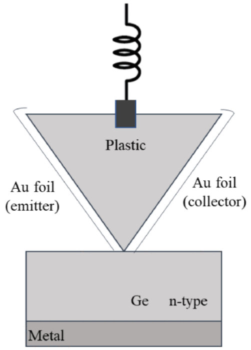

Figure FinFET.4: First point contact transistor.
Figure taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

Historical Development [2]

- **1987** — **First Multi-Gate Transistor**
- **1989** — **First Double-Gate Transistor named DELTA**
- **1997** — Fabrication of 30nm ultra-thin-body (UTB) SOI MOSFET device
- 1998 — Fabrication of 'folded channel transistor’
- **1999** — **Fabrication of** p-channel-based DG MOSFET structure
- **2001** — **Fabrication of first** quasi planar FinFET device

FinFET

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FinFET Structure and Operating Principle

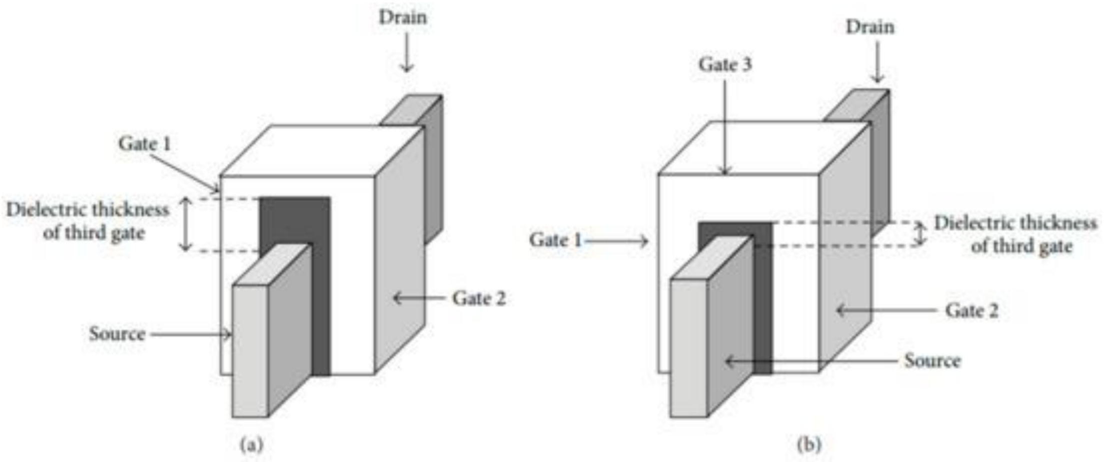


Figure FinFET.4: Schematic of (a) DG FinFET and (b) tri-gate FinFET. Figure taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

- Vertical "fin" channel with gate on three sides (tri-gate)
- H_{Fin} (fin height) controls drive current
- W_{Fin} (fin width) controls electrostatics
- Better channel control → lower leakage

FinFET Classifications

- **By substrate:** Bulk vs SOI FinFET (Fig. 5)
- **By gate type:** Short-Gate (3-terminal) vs Independent Gate (4-terminal) (Fig. 6)
- **By dielectric geometry:** Double-Gate vs Tri-Gate (Fig. 4)
- Tri-Gate = lower capacitance + better performance

Figure 4 reused for FinFET Classification.
Figures taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

Figure FinFET.5: Structure schematic of (a) Bulk FinFET and (b) SOI FinFET

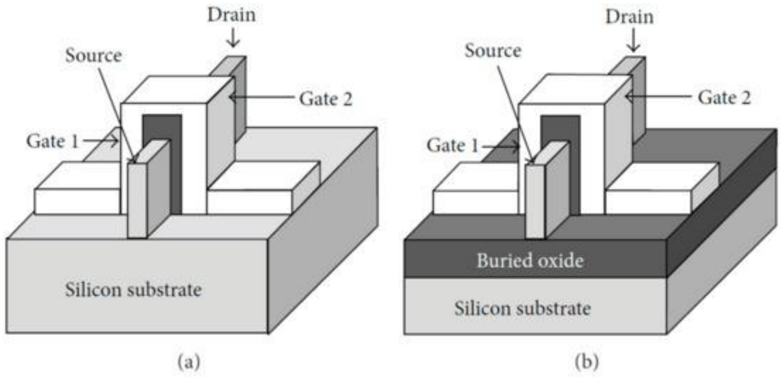


Figure FinFET.6: Structure schematic of (a) SG FinFET and (b) IG FinFET

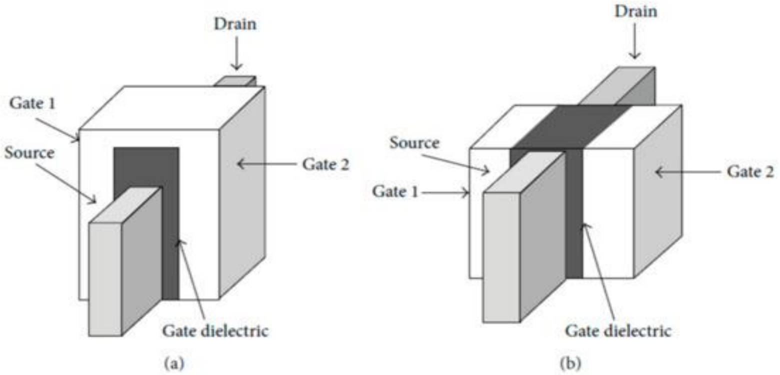
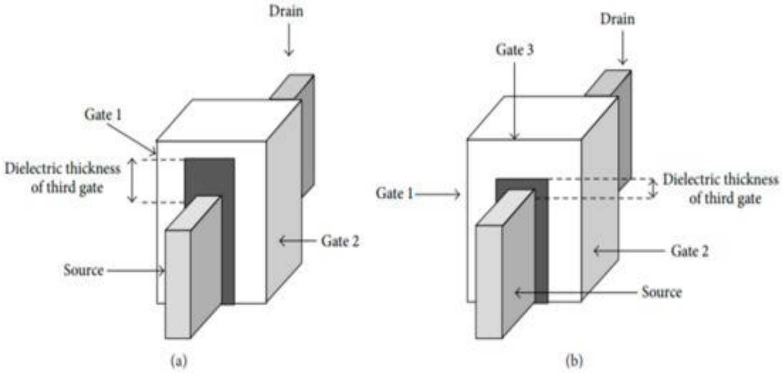


Figure FinFET.4: Schematic of (a) DG FinFET and (b) tri-gate FinFET.



Materials in FinFET Fabrication

Parameters (V_d = 0.75 V, V_g = 0.75 V)	(LaZrO2) Gate dielectric material k = 40	(SiO2) Gate dielectric material k = 3.9
I_{ON} (A)	4.95×10^{-5}	1.78×10^{-5}
I_{OFF} (A)	3.61×10^{-14}	5.02×10^{-13}
I_{ON}/I_{OFF}	1.37×10^{9}	3.50×10^{7}
V_t (V)	0.253	0.207
SS (mV/dec)	60.3	67.02
DIBL (mV/V)	10.1	43

Table FinFET.1: The impact of SiO2 and LaZrO2 gate dielectric material usage in n-FinFET device efficiency. Table taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

- **Gate dielectrics:** HfO_2, $LaZrO_2$ → high-k, low leakage
- **Channel materials:** Si, SiGe, GaAs, GaN → mobility ↑
- $LaZrO_2$ FinFET: Ion/Ioff ≈ 1.37×10^9 vs 3.5×10^7 for SiO_2
- HKMG stacks → better Vt stability

Key Challenges

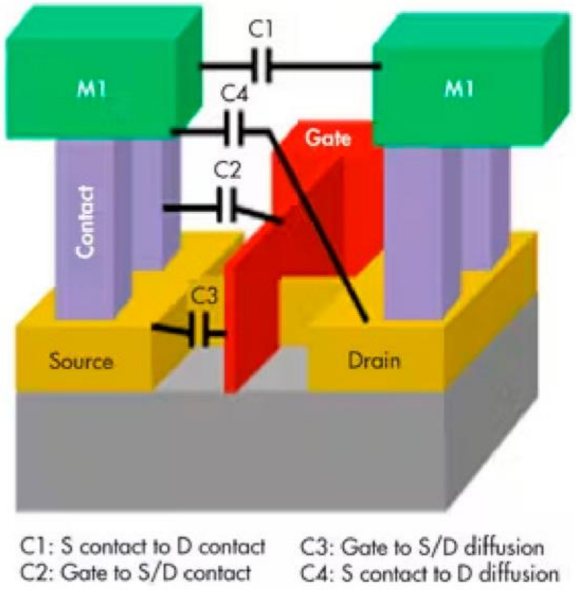


Figure FinFET.7: Parasitic capacitances of a FinFET

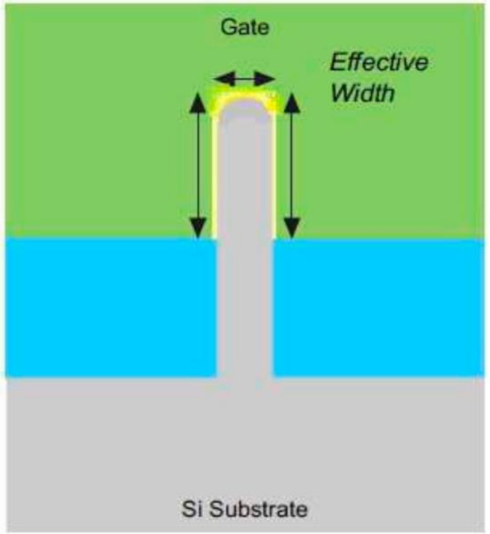


Figure FinFET.8: Cross-sectional view of a Tri-FinFET with curved fin corners

Figures taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

- Parasitic capacitances ↑ → slower switching
- Corner leakage → “corner effects”
- Quantum effects → mobility loss in ultra-thin fins
- Complex lithography and self-heating

Recent Innovations

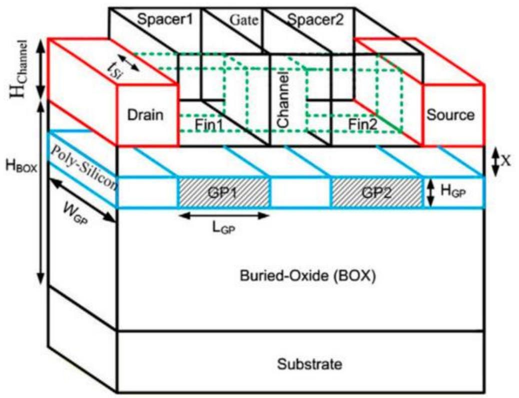


Figure FinFET.9: Three-dimensional view of the GP-FinFET structure schematic

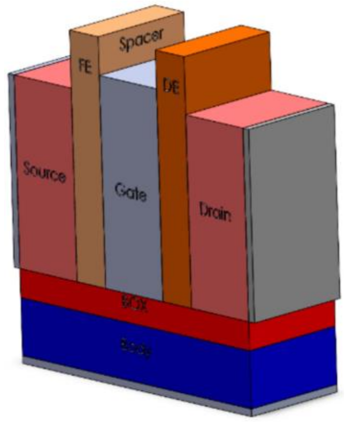


Figure FinFET.10: Three-dimensional view of the most effective structure of Negative Capacitance FinFET with Ferroelectric Spacer

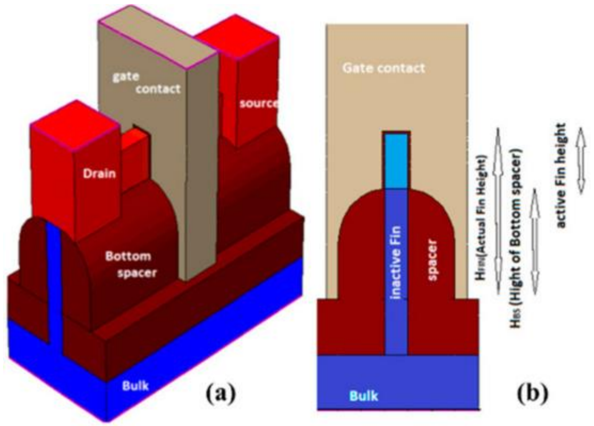


Figure FinFET.11: (**a**) 3D view and (**b**) side view of the structure of bottom spacer FinFET

Figures taken from Karimi, K.; Fardoost, A.;Javanmard, M. Comprehensive, Review of FinFET Technology: History, Structure, Challenges, Innovations, and Emerging Sensing Applications. Micromachines 2024, 15, 1187.

- **Ground-Plane FinFET (GP-FinFET):** lower DIBL
- **Bottom-Spacer FinFET:** better SCE & thermal control
- **Negative Capacitance FinFET:** ferroelectric spacer → higher Ion, lower SS

Conclusion

- FinFET = key bridge between planar CMOS and GAA
- Provides low power and scalability down to ≈ 5 nm
- Remaining issues: thermal management, manufacturing cost
- Next step → GAA (Multi-Bridge Channel FET)

FinFET

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Introduction to Gate-All-Around (GAA)

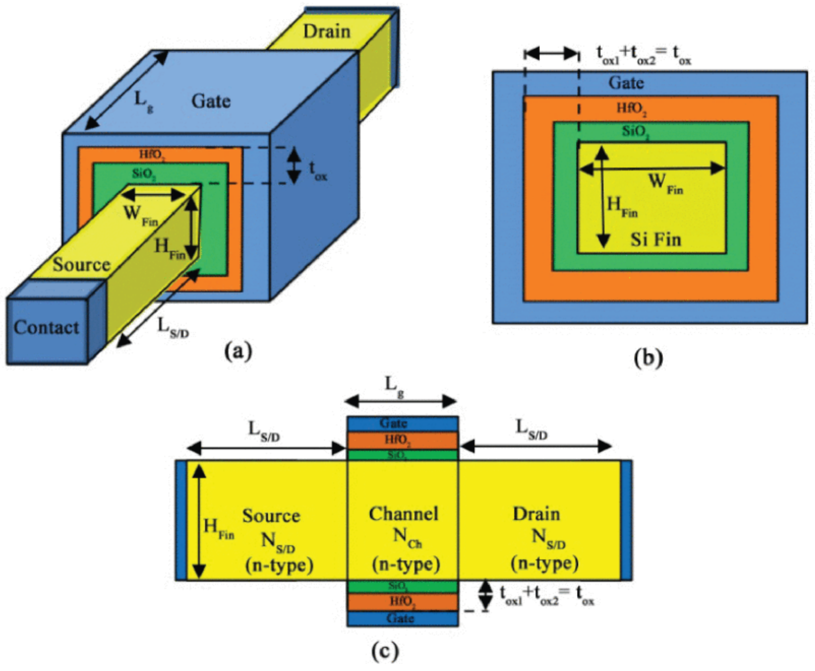


Figure FinFET.12: Schematic diagram of GAA structure along (a) Vertical and horizontal view (b) & (c). Figure taken from R. R. Das, T. R. Rajalekshmi and A. James, "FinFET to GAA MBCFET: A Review and Insights," in *IEEE Access*, vol. 12, pp. 50556-50577, 2024, doi: 10.1109/ACCESS.2024.3384428.

- GAA: channel fully surrounded by gate → 360° control
- Better suppression of SCE and DIBL
- Improved sub-threshold swing and mobility
- Compatible with existing FinFET fabrication flow

Motivation for GAA Evolution

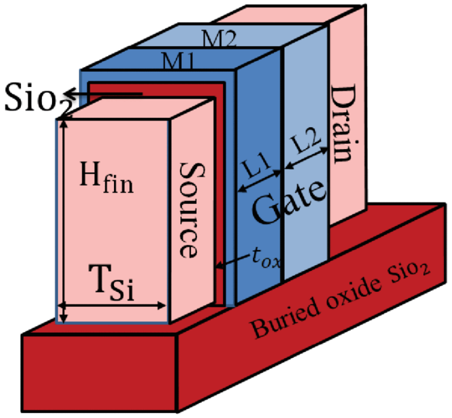


Figure FinFET.13: Schematic diagram of DMG FinFET

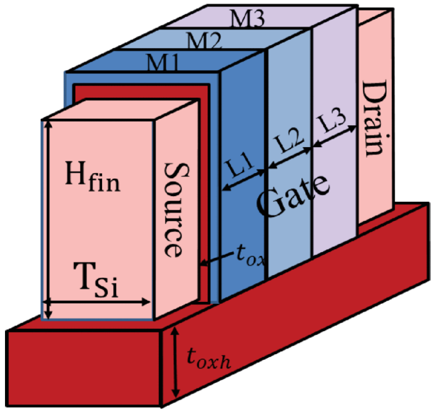


Figure FinFET.14: Schematic diagram of TMG FinFET

Figures taken from R. R. Das, T. R. Rajalekshmi and A. James, "FinFET to GAA MBCFET: A Review and Insights," in *IEEE Access*, vol. 12, pp. 50556-50577, 2024, doi: 10.1109/ACCESS.2024.3384428.

- FinFET scaling limited beyond 5 nm → electrostatics degrade
- Fin height & width quantization → reduced control
- GAA offers full gate wrapping → stronger channel coupling
- Enables continued Moore's Law scaling

From Nanowires to Nanosheets

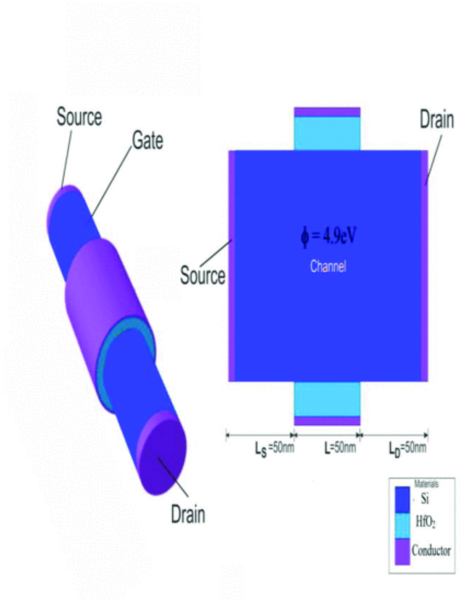


Figure FinFET.15: Cross Sectional and Schematic view of Nanowire FET. Figure taken from M. Singh, Harsh, M. Rani, P. Mittal, T. Chaudhary and B. Raj, "Nanowire FET Design and Analysis for Low-Power Applications," *2025 International Conference on Electronics, AI and Computing (EAIC)*, Jalandhar, India, 2025, pp. 1-5, doi: 10.1109/EAIC66483.2025.11101388.

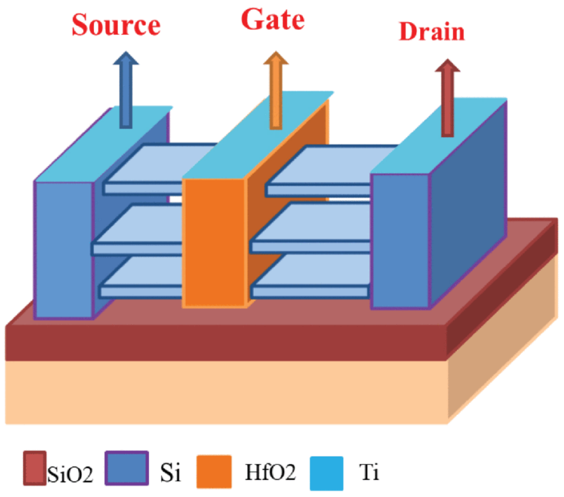


Figure FinFET.16: Cross Sectional and Schematic view of Nanowire FET. Figure taken from R. R. Das, T. R. Rajalekshmi and A. James, "FinFET to GAA MBCFET: A Review and Insights," in *IEEE Access*, vol. 12, pp. 50556-50577, 2024, doi: 10.1109/ACCESS.2024.3384428.

- Early GAAs used **nanowires** → excellent control but low drive current
- **Nanosheets (MBCFETs)** widen the channel → higher Ion
- Multi-Bridge structure = stacked nanosheets
- Samsung first adopted MBCFET for 3 nm node (2022)

FinFETs VS GAA MBCFETS

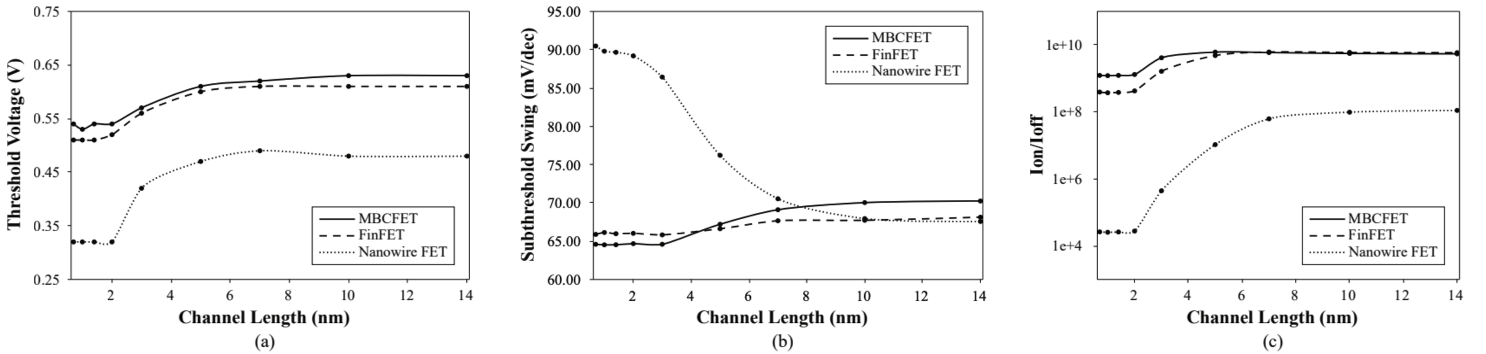
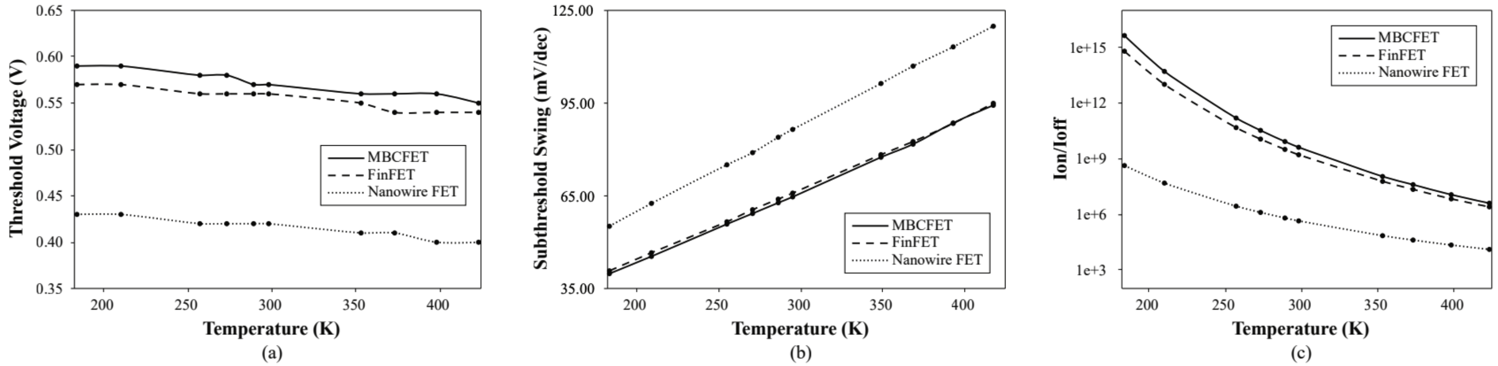


Figure FinFET.17.1: Effect of Channel Length on (a) Threshold Voltage (b) Subthreshold Swing (c) Ion/Ioff ratio.

Parameter	Conventional	Optimized
Threshold Voltage	0.49V	0.57V
Subthreshold Swing	73.02mV	64.60mV
I_{on}/I_{off} ratio	2.81e+7	4.13e+9

Figure FinFET.17.2: Performance comparison.



Figures FinFET.18.1: Effect of Temperature on (a) Threshold Voltage (b) Subthreshold Swing (c) Ion/Ioff ratio.

Parameter	Nanowire FET	FinFET	MBCFET
Threshold Voltage	0.42V	0.56V	0.57V
Subthreshold Swing	86.43mV	65.83mV	64.60mV
I_{on}/I_{off} ratio	4.44e+5	1.63e+9	4.13e+9
Perfromance Degradation for 80°C	14.14%	15.56%	16.37%

Figures FinFET.18.2: Performance comparison.

Figures taken from Ahmed, Foez & Paul, Robi & Saha, J.. (2020). Comparative Performance Analysis of TMD based Multi-Bridge Channel Field Effect Transistor. 10.1109/NAP51477.2020.9309688 [9]

FinFETs VS GAA MBCFETS

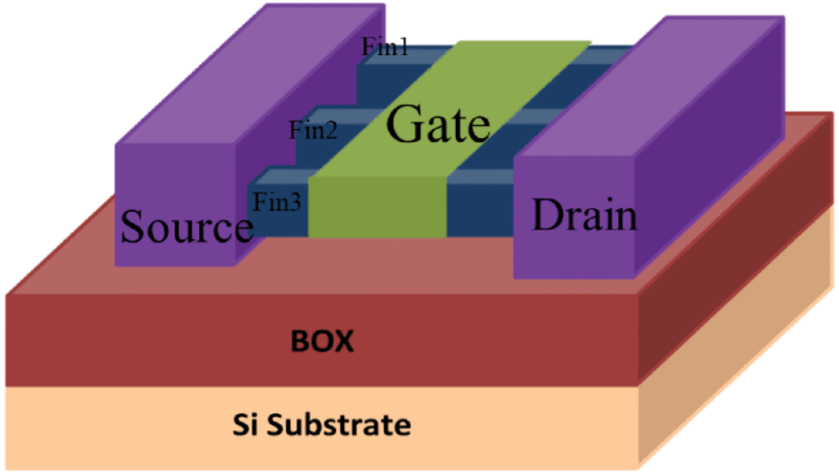


Figure FinFET.19: Schematic diagram of M-FinFET. Figure taken from R. R. Das, T. R. Rajalekshmi and A. James, "FinFET to GAA MBCFET: A Review and Insights," in *IEEE Access*, vol. 12, pp. 50556-50577, 2024, doi: 10.1109/ACCESS.2024.3384428.

- **FOOTPRINT AREA AND SPEED**
  - M-FinFET ↑ area/fin, GAA MBCFET ↑channels, = area.
- **INTERNAL STRUCTURE**
  - Dynamically alter the channel width, which is a functionality absent in traditional FinFETs
- **PRODUCTION**
  - Identical process tools & manufacturing procedures →Seamless integration
- **LOW LEAKAGE CURRENTS, OPERATIONAL VOLTAGE, AND DYNAMIC POWER**
  - GAA →electrostatic gate control capacity ↑, leakage ↓
  - GAAFETs →scalability & gate control, FinFETs →low power
  - Combination: Voltage scaling and adaptive body biasing

Where the Industry Stands Today

Company	Current Node (2024-2025)	Next-Gen Target	Technology in Use	Adoption Timeline
Samsung	3 nm (3GAE / 3GAP) – world's first GAA MBCFET chips (since 2022)[10]	2 nm GAA MBCFET	**GAA MBCFET (3 nm)** + FinFET legacy nodes	First gen GAA → 2022 2nd gen → 2023 2 nm planned 2025
TSMC	3 nm FinFET (N3 series) [11]	2 nm GAA Nanosheet (N2)	**FinFET**	Developing GAA for ~2026 launch
Intel	Intel 4 / Intel 3 FinFET	2 nm "Intel 20A" = **RibbonFET (GAA) [12]**	**FinFET**	GAA introduction ~2024 (20A node)

FD-SOI

- Technology overview
- “A Novel 1T-APS Using 22nm FD-SOI Technology With Record High Tunable Sensitivity Range”, H. Xie et al. [3]
- “A Wideband Sub-6GHz Transceiver Front End for 5G Base Stations in 22-nm FD-SOI CMOS”, C. Elgaard et al. [4]

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Fully Depleted-Silicon On Insulator: Overview

- Thin Si channel depleted of free carriers
- Back-gate V_{th} control [5][6][8]
- Reduced [5]:
  - Junction capacitance
  - Leakage current
  - Power consumption
  - Output conductance g_{ds}
- Increased:
  - Transconductance g_m [5]
  - Intrinsic voltage gain $A_v = -{g_m}/{g_{\text{ds}}}$

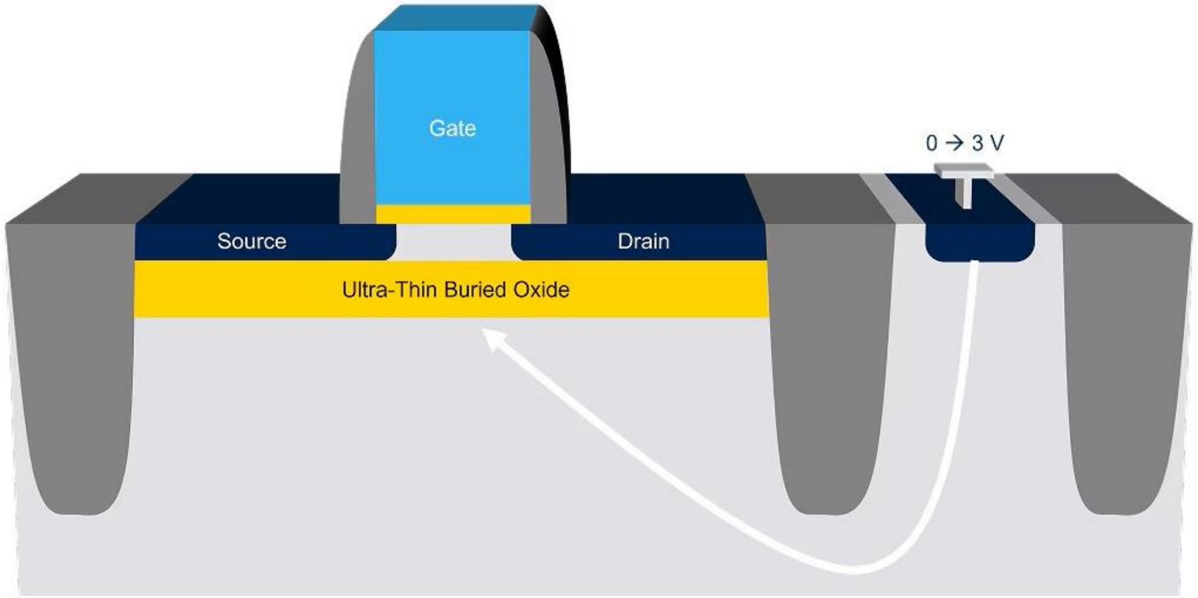

Figure FD-SOI.1: 3D Render of FD-SOI transistor. Taken from STMicroelectronics.

Fully Depleted-Silicon On Insulator: Overview

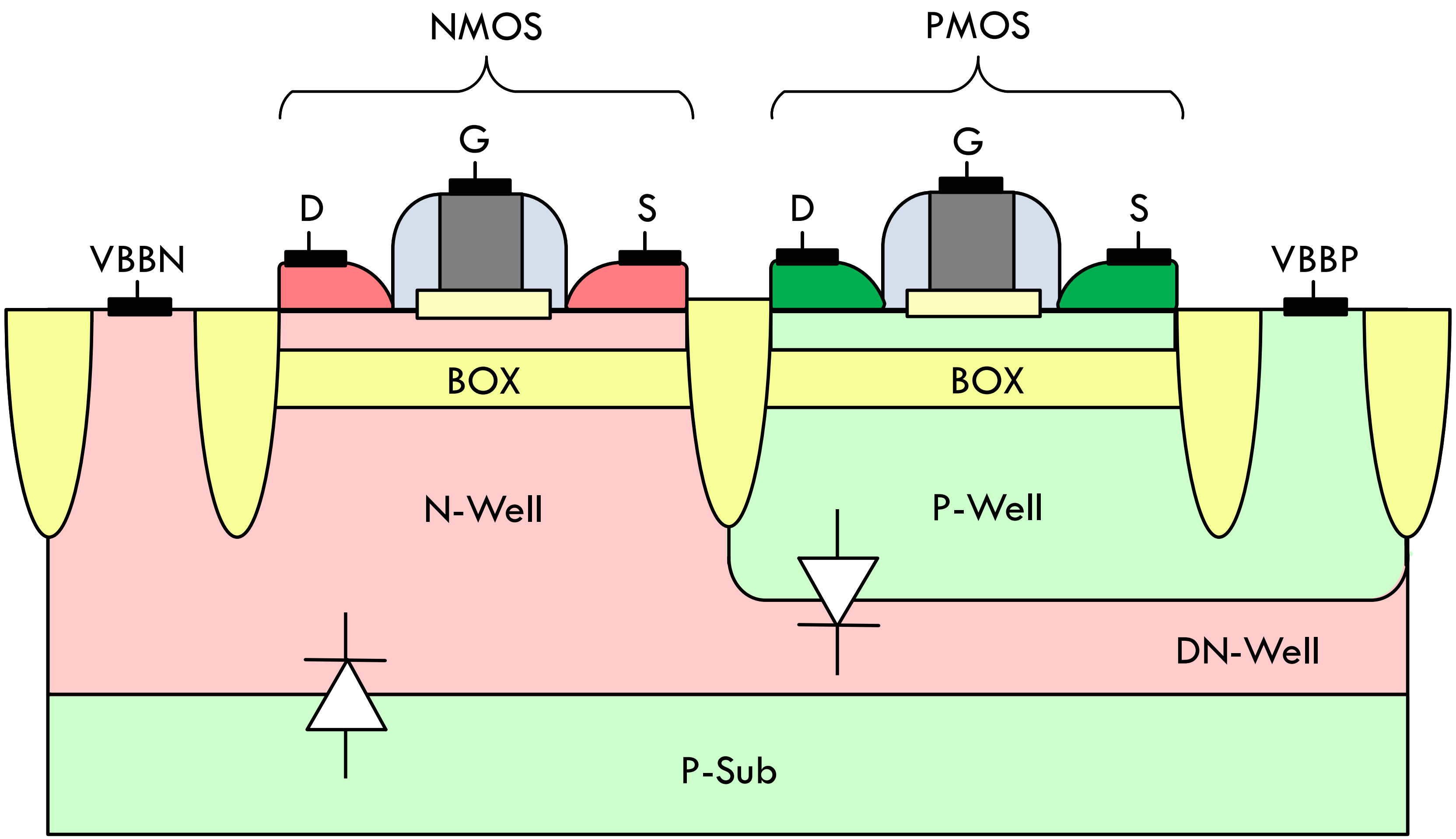

Figure FD-SOI.2: Low threshold voltage (LVT) FD-SOI CMOS. Figure taken from Patrick Scheer, *CMOS technologies modeling for RF/millimeter-Wave applications*, Grenobe INP, Phelma, Dec. 2024.

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1T-APS Using 22nm FD-SOI Technology

Figure FD-SOI.3: Typical CMOS 4T-APS (four-transistor active pixel sensor). Taken from [7].

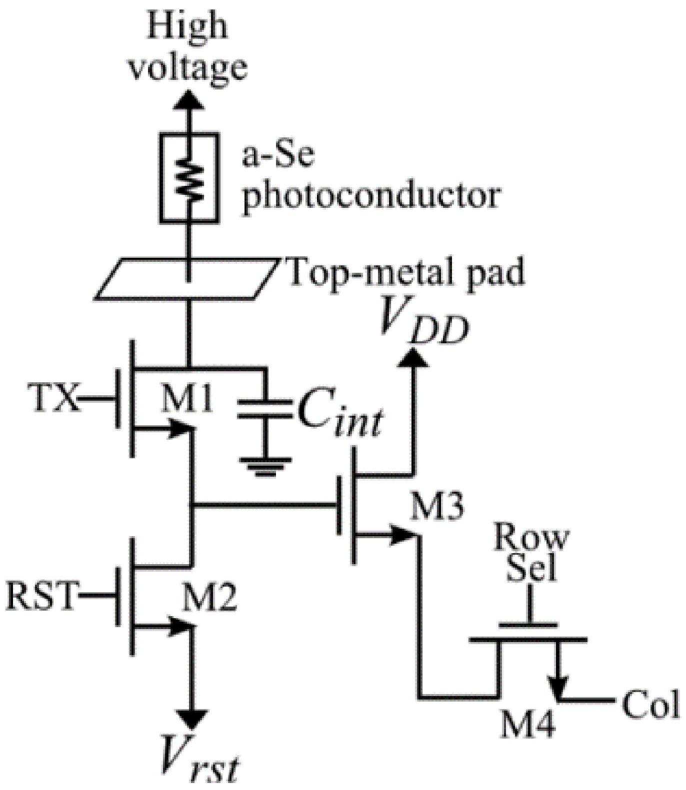
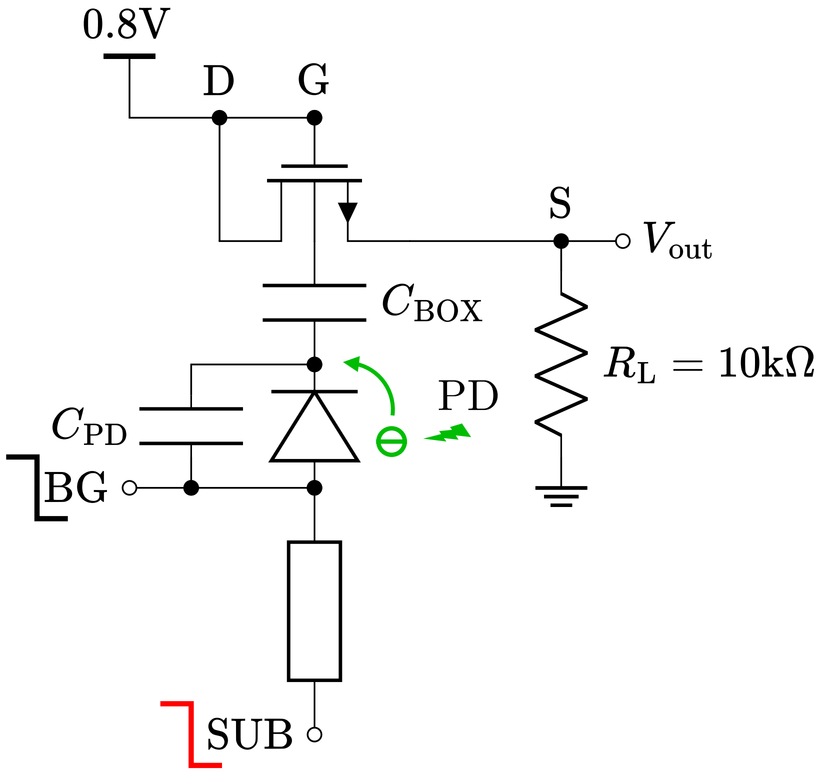


Figure FD-SOI.4: Equivalent circuit of the 1T-APS in FD-SOI 22nm proposed by [3].



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[10] https://semiconductor.samsung.com/processor/mobile-processor/

[11] https://www.tsmc.com/english/dedicatedFoundry/technology/logic

[12] https://www.intel.com/content/www/us/en/foundry/process.html